

EE6008 GUIDED READING EDITION

Investigation of Communicative Flight Paths for Small Unmanned Aerial Systems

Original paper interleaved with English section-by-section guidance

BIG PICTURE. This is a human-robot interaction study of whether novice observers attach common meanings to eight large, biologically inspired UAV paths. It studies semantic interpretation, not digital or covert communication.

ROLE IN YOUR PROJECT. Use its trajectories as a conspicuous positive control and shape library. Spiral-to-Landing can support an overt public-safety layer; the final keyed digital codebook should use much smaller mission-embedded deviations.

How to use this edition

- Read original pages normally; blue commentary appears after the nearest page boundary for each major section.
- Use Purpose and the numbered process first, then revisit equations and figures with the vocabulary table.
- Treat red Critical Reading boxes as assumptions or evidence gaps to verify; use transfer notes for your EE6008 plan.

PLACEMENT NOTE. IEEE sections often share a source page: one section may end and the next may begin on the same page. Guide pages are therefore inserted after the closest complete source page and identify the exact section they explain.

Guide map

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SOURCE CITATION. B. A. Duncan, E. Beachly, A. Bevins, S. G. Elbaum, and C. Detweiler, IEEE ICRA 2018, pp. 602-609. DOI: 10.1109/ICRA.2018.8462871.

Investigation of Communicative Flight Paths for small Unmanned Aerial Systems*

Brittany A. Duncan¹, Evan Beachly¹, Alisha Bevins¹, Sebasitan Elbaum¹, and Carrick Detweiler¹

Abstract—This project seeks to generate small Unmanned Aerial System (sUAS) flight paths that are broadly understood by the general population and can communicate states about both the sUAS and its understanding of the world. Previous work in sUAS flight paths has sought to communicate intent, destination, or emotion of the system without focusing on concrete states (e.g., low battery, landing, etc.). This work leverages biologically-based flight paths and experimental methodologies from human-human and human-humanoid robot interactions to assess the understanding of avian flight paths to communicate sUAS states to novice users. If successful, this work should inform: the human-robot interaction community about the perception of flight paths, sUAS manufacturers on how their systems could communicate with both operators and bystanders, and end users on ways to communicate with others when flying systems in public spaces. General design implications and future directions of work are suggested to build on the results here, which suggest that novice users gravitate towards labels they understand (draw attention and landing) while avoiding more technical labels (lost sensor).

I. INTRODUCTION

Small unmanned aerial systems (sUAS) have sophisticated control stations and a rich variety of interfaces to communicate with their operators. Yet, as these vehicles become part of applications involving stakeholders that are not the operators, they will increasingly need to establish broader communication channels in order to be accepted in public spaces and to create safer interactions.

Consider for example existing public-facing applications, such as Amazon Prime Air [1] or the Alphabet’s burrito delivery [2]. For such applications, the sUAS might be required to communicate not just with their control base, but also with the customers expecting a delivery, and bystanders that may not be fully aware of the intent of the vehicle. These other stakeholders may not have experience dealing with the sUAS but they will ultimately render their judgment about the application in part based on how the vehicle communicates with them. In other more specialized application contexts, like that of sUAS supporting fire management activities [3] as depicted in Fig. 1, communicating dangerous situations to fire personnel and accidental observers is critical. Alternately, these gestures could be used in agricultural applications of UAS, such as those in orchards [4] where most workers are unlikely to be trained on technologies unrelated to their tasks, but should be alerted to off-nominal operations.

*This work was supported by NSF NRI 1638099

¹All authors are affiliated with the Department of Computer Science & Engineering, University of Nebraska, Lincoln, NE 68588, USA {bduncan, ebeachly, abevins, elbaum, cdetweiler}@cse.unl.edu



Fig. 1. Concept imagery depicting UAV operators at a prescribed fire, where bystanders work around the fire and may need state information.

Our approach to communicating to such stakeholders is through sUAS gestures, more specifically flight motions that convey the sUAS state. This communication medium is appealing in that it requires no additional equipment (such as speakers or lights) and can be easily incorporated into existing systems. If well designed, these motions are also robust to communication challenges such as partial occlusion, viewing angle, or ambient lighting (as in [5]).

One of the challenges of using sUAS gestures is identifying those that can be consistently interpreted by stakeholders that may not have been trained in the technology. In this work we start investigating this challenge by asking: *Do novice users show broad agreement on the meaning of sUAS gestures?* We investigate this question using Amazon’s Mechanical Turk (mTurk) platform to gain access to 64 general users for a video-based study of sUAS communications.

Leveraging methods from human gesture understanding, this paper contributes the first study of general communications by sUAS. It is distinct from previous work on sUAS gestural communication in that we are attempting to communicate relevant action information from simple sUAS gestures to novice users, instead of attempting to mirror users’ emotions or communicate about only direction of flight. The results indicate that novice participants are able to properly label gestures associated with landing or drawing attention, with less agreement for other gestures. We also found that, contrary to the findings of previous work, user attitudes towards robots did not seem to affect their ability to recognize the meaning of a gesture.

I. INTRODUCTION | INSERTED AFTER SOURCE PAGE 1

A human-robot interaction question, not a digital channel

PURPOSE. This section asks whether untrained bystanders attach common meanings to UAV flight gestures. The goal is public understanding and safety, not covert bit transmission.

What this section is doing

- 1 Identify public-facing situations where people near a drone may not share the operator's interface or training.
- 2 Propose the flight path itself as a gesture that communicates the vehicle's internal state or understanding of the world.
- 3 Ask whether novice observers show broad agreement when assigning labels to those gestures.
- 4 Preview a 64-person video study and the finding that landing and attention meanings are easier than technical fault states.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
sUAS	Small Unmanned Aerial System, including the aircraft and supporting system.
Bystander	A person near the operation who is not necessarily the pilot or intended user.
Communicative gesture	A movement deliberately designed to convey a meaning to an observer.
Semantic label	The human-readable meaning assigned to a gesture, such as landing or lost sensor.
Novice user	A participant without assumed specialist knowledge of the system.
HRI	Human-Robot Interaction: the study of how people perceive, understand, and work with robots.
Off-nominal state	A condition outside expected normal operation, such as sensor or signal loss.

How to read the equations, figures, or evidence

- The transmitter is the whole aircraft, and the receiver is an untrained human watching a video.
- The dependent outcome is agreement on meaning, not BER, throughput, timing recovery, or machine classification.
- The paper is intentionally searching for salient, understandable motion; this objective points in the opposite direction from covert micro-motion.

What transfers to the EE6008 project

- Use this paper as a conspicuous positive-control baseline: a good Willie should detect these deliberate gestures.
- Use stable public meanings, especially spiral-to-landing, only for an overt safety layer separate from the keyed digital layer.
- Do not include human participants in the MVP unless ethics approval and supervisor direction are obtained.

CRITICAL READING. The paper's claim that biologically inspired motion may tolerate viewing angle or occlusion is a motivation, not a result of this experiment. The actual study uses fixed videos and a controlled viewpoint.

Check your understanding

- Who is the intended receiver in this paper?
- Why is a gesture that is easy for bystanders to understand unsuitable as a covert symbol?

II. RELATED WORK

The human ability to infer intentionality from random motion has been well established, beginning with Heider's work on apparent behavior in 1944 [6], which was later extended to understand perception of biological motions from humans and animals [7], [8]. Through studies such as these, we can begin to understand the intentionality that is applied to observed motion and the components that make this intentionality more broadly understood. In this section, relevant work on human and robot gestural communications will be presented.

A. Human Gestural Communications

Human gestural communications have been studied for their communicative ability in order to understand how they are perceived and what they can be used to communicate. Krauss, Morrel-Samuels, and Colasante [9] conducted a set of studies to understand how co-speech hand gestures are understood and found that while hand gestures convey some information, they do not communicate as well as speech. Prati and Pietrantoni [10] investigated the use of hand gestures when verbal communication would be difficult to understand differences in communicative ability of different gesture types. Their participants watched videos of firefighters performing ten gestures and labeled them using free response. In both studies, these gestures had meanings that were similar to gestures participants had previously seen.

B. Robot Gestural Communications

Gestural communications in robots can be split into ground robot gestural communications and sUAS gestural communications. While gestures have been examined in humanoid robots, this has been limited to social gestures and collaborative gestures. The current state of the art with sUAS has been to communicate high-level state information or to use gestures for control of a vehicle.

1) *Ground Robot Gestural Communications*: Social gestures have been investigated in HRI in much the same way that was described above for communicative hand gestures. Salem et al. [11] investigated the ability for co-speech gestures to enhance humanoid robot communications. Huang and Mutlu [12] evaluated the use of gestures to improve recall in humanoid robot interactions. Ng, Luo, and Okita [13] developed a gesture model to produce gestures from text input and tested the modification of parameters to convey excitement or expressiveness. Riek et al. [14] tested cooperative social gestures on a humanoid robot to understand the impact of speed and viewing angle, and found that negative attitudes towards robots correlated with a decreased ability to understand the gestures in the study. Overall, these works have assessed understanding of gestures, but they are focused on leveraging the existing understanding of participants from interacting with other people in order to improve humanoid robot communications.

Of more interest to this work are the collaborative gestures that have been developed primarily for industrial applications as in [15], [16], but one limitation of the work in this area

is the assumed presence of a visible goal as reported in [17]. Dragan and Srinivasa [15] tested the integration of an observer into motion planning for an industrial robot. Gleeson et al. [16] observed gestural communications between humans, derived terms and gestures for use by the robot, and implemented them on a robot to observe their communicative ability. Both of these studies indicated that gestures were more effective when they conveyed context and goal, which is a challenge for the sUAS gestures to overcome.

2) *sUAS Gestural Communications*: Communications with sUAS can be split into communication from the sUAS and communication to the sUAS.

a) *Communication from sUAS*: Communicative flight paths have been investigated for ability to communicate affective state [18], [19], intended destination [20], and intended direction of flight [5]. These flight paths would enhance interaction with sUAS in collocated environments, but do not communicate actions or states that might be necessary in uses with more bystanders or broader application.

Sharma et al. [18] investigated the ability to communicate affect via flight path with collocated users and found that to increase valence or arousal communication, space should be used more indirectly and the motions should be faster. Cauchard et al. [19] explored personality models for the sUAS to increase interest in interaction and possibly allow them to mirror the personality of their users in the future. Findings from both studies were operationalized by keeping these parameters as constant as possible across flight paths.

Szafir, Mutlu, and Fong [20] used both mTurk and in-person interactions to explore the perception of animation principles applied to sUAS flight paths to increase the communication of intent. Szafir, Mutlu, and Fong [5] next assessed the ability of a light ring to communicate the direction of sUAS flight through in-person testing where the participants would predict the end state of the vehicle. This work considered: viewing angles, movement in multiple dimensions, occlusion, and ambient lighting. While very informative to the field at large, this work is better applied to close interactions in more controlled environments so we primarily focused on possible problems in communication.

b) *Communication to sUAS*: Gestural communication to sUAS for commanded control has been investigated [21], [22], [23], but this work is not directly relevant to the work described here.

III. DEFINING sUAS COMMUNICATIVE FLIGHT PATHS

This paper presents an initial study to address the question: *Do novice users show broad agreement on the meaning of sUAS gestures?* From a methodology perspective we start exploring this question by following established protocols used to investigate human gestures [9], which seek to understand the level of agreement by exposing participants to a limited gesture set and then requesting those participants to apply a label from a limited set. From the sUAS gesture perspective, we start by adopting flight paths used in nature, which are robust to viewing angle or occlusion, oscillatory

II. RELATED WORK | INSERTED AFTER SOURCE PAGE 2

Where communicative flight paths come from

PURPOSE. This section builds the conceptual bridge from human gesture and humanoid-robot gesture to aerial motion that communicates affect, intent, direction, or state.

What this section is doing

- 1 Establish that humans infer intentional behaviour from simple moving shapes and biological motion.
- 2 Review human co-speech and emergency gestures, where prior familiarity strongly affects interpretation.
- 3 Review humanoid and industrial robot gestures, emphasising the roles of context and visible goals.
- 4 Separate communication from the sUAS to people from gesture commands sent by people to the sUAS.
- 5 Identify the gap: previous drone motion mostly communicates affect, destination, or direction rather than concrete operational states.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Intentionality	The observer's belief that a motion is purposeful rather than random.
Biological motion	Movement patterns characteristic of living agents and often recognised from sparse cues.
Co-speech gesture	A hand or body gesture produced together with spoken language.
Affect	Emotion-related qualities such as pleasantness and activation.
Valence / arousal	How positive or negative an emotion is, and how energetic or intense it is.
Collocated interaction	Human and robot occupy the same physical environment.
Mental model	A person's internal expectation of how a system behaves and what its actions mean.

How to read the equations, figures, or evidence

- Watch how prior context changes interpretation: a gesture can become understandable because it resembles a familiar human action or because a visible goal reveals the intent.
- The authors deliberately remove a visible goal later, so participants must judge the path shape rather than infer an obvious destination.
- The literature motivates semantic understanding; it does not provide a digital protocol, code rate, synchronisation method, or secrecy model.

What transfers to the EE6008 project

- Treat semantic gesture literature as a source of human-salience features that Willie may exploit.
- Separate overt intent communication from covert machine communication in the system architecture and report.
- When selecting normal-flight H0 data, include realistic goal-directed turns and corrections so Willie cannot win by detecting any curvature at all.

CRITICAL READING. Meanings learned from familiar human gestures or a specific culture may not generalise to different countries, platforms, tasks, or viewing geometries.

Check your understanding

- What earlier drone-gesture meanings were studied before concrete system states?
- How can a visible goal act as a confounding cue?

in nature to allow looping, and adapted from biological inspiration to explore any templates that might exist. Given the formative stage of the work, we limit the impact of environmental factors (through being performed in an indoor space), constrain the labels (to understand agreement rather than generation), and do not introduce a visible goal state (to assess understanding rather than inference). Further description of the motions and labels are described in Section IV and the methodology details appear in Section V.

IV. sUAS FLIGHT PATH DESIGN

The initial paths created from this work were developed from flight patterns used by birds in order to leverage the advantages inherent in biologically inspired behaviors, as described in [24], [25]. This section will describe the available labels, flight path selection, programming environment, and video creation for the experiments described later in this work.

A. Possible Flight Path Labels

The current labels were chosen based on likelihood that they would be encountered in flights and generally would require redirection or intervention from the operator, or awareness from bystanders. It was also anticipated that these states would be understood by novice users due to the widespread use of hobbyist systems or observations of other aircraft (e.g., Landing, Low Battery, Draw Attention), commonality with other taskable systems (e.g., Missed Goal, Change Position), and potential similarities to states encountered in smart phone technology (e.g., Lost Sensor, Lost Signal). Another consideration was to choose states that were domain independent rather than focusing on possible applications of the technology (e.g., not Deploying Sensors nor Taking Pictures).

B. Flight Path Selection

The avian flight paths we selected were originally identified by Davis [7] as oscillatory motions (those with a steady periodic motion and which could be created from sinusoid functions). More details on their inclusion/exclusion criteria can be found in the original work, but these motions were of interest to this work because they are biologically inspired, can be created in a replicable way, offer the ability to scale and loop as needed, and can generally be perceived in the presence of occlusion or multiple viewing angles. The requirement for biologically inspired behaviors also takes into consideration the requirements for deployment of these motions, such as the need to be observable against a natural background, able to contend with energy constraints, and understandable by other animals (or in this case humans). The eight cyclic motions used by birds and identified in [7] are: Circle, Figure-8, Left-Right, Loop, Spiral, Swoop, Undulate, and Up-Down

When designing the labels for this study, we considered states that may impact and may need to be communicated to bystanders. The states we chose were: lost signal, lost sensor, draw attention, landing, missed goal, change position,

and low battery. We then performed an initial assignment of those labels to the motions to later gauge whether participants would confirm these assignments or realize alternative ones. The initial assignments with a brief description of the thought behind these assignments follows:

- Circle: lost signal, in which the movement could help the sUAS regain signal
- Figure-8: lost sensor, which looks like the motion used to recalibrate your phone's magnetometer
- Left-Right: missed goal, which looks like shaking head
- Loop: draw attention, which might be reminiscent of a ferris wheel
- Spiral: landing, which could be used in indicating a position of landing
- Swoop: draw attention, since this is eye catching
- Undulate: change position, since this motion could be performed while starting in the direction
- Up-Down: change position, which looks like nodding to acknowledge the command

C. Flight Path Programming

To perform each gesture, the Ascending Technologies Firefly hexcopter (weighing less than 1.6 kg and 60.5x66.5 cm) would take-off, hover at the starting point of the gesture, perform the flight path representing the gesture for 30 seconds, and then land.

To create the flight paths, the sUAS autonomously flew along pre-programmed paths. A Vicon motion capture system tracked reflective markers attached to the sUAS and measured its position and orientation to a high degree of accuracy (submillimeter error and 200Hz). The sUAS performed each gesture by following a target position that was continuously moved through a three dimensional space, using a PID pose controller to have the sUAS chase the target. The target's x , y , and z coordinates were coded as mathematical functions of time, as described next, thus yielding a parametric path for the gesture. Each gesture was programmed to move the sUAS at approximately 1 meter per second.

1) *Flight Path Parameters*: Table I shows the equations used to generate the eight paths. The x , y , and z positions are evaluated based on a specified range of time, t , starting at zero. In addition, the table shows the typical displacements observed during the sUAS flights for each motion.

2) *Flight Path Visualizations*: Fig. 2 shows a visualization of some of the flight paths reproduced from logged pose data. The red path shows the actual sUAS path, each square is 1 meter wide, and the gesture primarily took place over the center of the grid. Some gestures (e.g., swoop) are shown from an off-angle so that the path can be better viewed.

D. Flight Path Video Creation

Each flight was filmed by a video camera, and the video was later trimmed to include only the gesture and to ensure that all gesture videos were the same length and size. The camera's view was orthogonal to the gestures' width and height displacements, and its height centered the gestures within its view. Each video was uploaded to Youtube.com

III. DEFINING SUAS COMMUNICATIVE FLIGHT PATHS | INSERTED AFTER SOURCE PAGE 3

Turning a broad question into a controlled formative study

PURPOSE. This short section fixes the experimental philosophy: start with repeatable oscillatory paths and constrained labels under controlled indoor conditions, then ask whether opinions converge.

What this section is doing

- 1 Select flight motions inspired by nature because periodic patterns can be looped, scaled, and reproduced mathematically.
- 2 Constrain the environment to indoor video and limit the response labels.
- 3 Remove a visible goal state so participants label the motion itself rather than infer the task destination.
- 4 Treat the study as formative: it is an initial sanity check before richer open-ended or live experiments.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Formative study	An early experiment used to shape later design rather than deliver a final generalisable answer.
Oscillatory motion	A repeating or periodic movement, often generated with sine and cosine functions.
Controlled variable	A factor held constant to reduce alternative explanations for an outcome.
Response set	The labels or answers a participant is allowed to choose.
Confound	An uncontrolled factor that could explain the observed result instead of the intended cause.
Sanity check	A limited test that verifies an idea is plausible before investing in a larger study.

How to read the equations, figures, or evidence

- The design deliberately trades ecological realism for interpretability and repeatability.
- Robustness to viewing angle, occlusion, and outdoor background is an aspiration inherited from the biological inspiration, not tested here.

What transfers to the EE6008 project

- Adopt the same baseline-first logic: validate the receiver on deliberately large symbols before shrinking to micro-motion.
- Keep one factor sweep at a time during early experiments, but later randomise mission paths and environmental conditions.
- Document which conclusions apply only to the controlled indoor setup.

CRITICAL READING. A constrained label set can manufacture apparent agreement by preventing participants from expressing an unlisted interpretation. The later 7AFC condition reduces but does not remove this issue.

Check your understanding

- Which factors are controlled, and which real-world factors are postponed?
- Why choose periodic paths for an initial study?

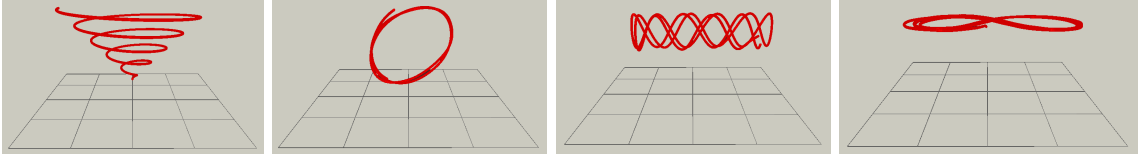


Fig. 2. Visualizations of the Spiral, Loop, Undulate, and Figure-8 paths from the flight log. Each square is 1m wide.

TABLE I
MOTIONS' EQUATIONS AND TYPICAL DISPLACEMENTS OBSERVED.

Motion	Equations	Displacement (m)
Circle	for $0 \leq t \leq 30$: $x = 2.5\cos(t)$ $y = 2.5\sin(t)$ $z = 1.5$	$x : 2.5m$ $y : 2.5m$ $z : 0.0m$
Figure-8	for $0 \leq t < 2PI$: $x = 0.75 - 0.75\cos(t)$ $y = 0.75\sin(t)$ $z = 1.5$ for $2PI \leq t < 4PI$: $x = -0.75 + 0.75\cos(t)$ $y = 0.75\sin(t)$ $z = 1.5$	$x : 4.0m$ $y : 2.0m$ $z : 0.0m$
Left-Right	for $0 \leq t \leq 30$: $x = 1.5\sin(1.5t)$ $y = 0$ $z = 1.5$	$x : 2.5m$ $y : 0.0m$ $z : 0.0m$
Loop	for $0 \leq t \leq 30$: $x = 0.75\sin(0.75t)$ $y = 0$ $z = 1.25$ $+0.9 \times 0.75\cos(0.75t)$ $+0.4 \times 0.75\sin(0.75t)$	$x : 2.0m$ $y : 0.0m$ $z : 1.5m$
Spiral	for $0 \leq t \leq 30$: $x = \cos(t) \times (30 - t)/20$ $y = \sin(t) \times (30 - t)/20$ $z = 2 - t/20$	$x : 2.0m$ $y : 2.0m$ $z : 1.5m$
Swoop	for $0 \leq t < 2.4$: $x = 0.5t$ $y = 0$ $z = 0.5 + ((t - 1.2)^2)$ for $2.4 \leq t \leq 4.8$: $x = 1.2 - 0.5t$ $y = 0$ $z = 0.5 + ((t - 1.2)^2)$	$x : 1.5m$ $y : 0.0m$ $z : 1.5m$
Undulate	for $0 \leq t \leq 6$: $x = -1.5 + t/2$ $y = 0$ $z = 1.5 + 0.5\sin(3.14 * t)$ for $6 \leq t \leq 12$: $x = 1.5 - t/2$ $y = 0$ $z = 1.5 + 0.5\sin(3.14 * t)$	$x : 3.0m$ $y : 0.0m$ $z : 0.8m$
Up-Down	for $0 \leq t \leq 15$: $x = 0$ $y = 0$ $z = 1.25 + 0.4\sin(4t)$	$x : 0.0m$ $y : 0.0m$ $z : 0.5m$

with dimensions of 854 by 480 pixels and sound was included in each video. Original videos can be found at <https://unl.box.com/v/ICRA-duncan-videos> and a summary of representative videos is included in the media attachment to this paper.

V. FORCED CHOICE LABELING STUDY

This study investigates the ability of novice users to discern possible intent of gestures. We employ the n-alternative forced-choice technique, which has been used in psychology [26], [27], [28], human-robot interactions [29], [30], and human gesture recognition in [9], [10]. More specifically, we conduct two studies. The first one employs a two-alternative

forced-choice (2AFC) model to test the recognition of a gesture from two labels (one chosen by us and one distractor). The second one, a seven-alternative forced-choice (7AFC), is meant to test the recognition of a gesture among seven labels, to assess broad agreement.

A. Approach

As in [9], it was expected that for the 2AFC condition users would be able to select the preferred label (chosen by the experimenters) more often than chance. One major difference in this work than in the work on co-speech gestures by [9], [10], [12], [11], [13] is that the gestures tested here do not have an inherent meaning within speech; instead we are assessing whether they map to concepts assumed to be already understood by the users.

We also wanted to take this experiment a step further to see if there was convergence from participants on label assignment by testing all labels and all gestures in the 7AFC task. Here, it was expected that gestures with high recognition in the 2AFC task would also show high convergence in the 7AFC condition.

B. Participants

Participants were a convenience group recruited from mTurk and paid \$2.00 dollars for their participation in the experiment. All participants were required to have a 95% task approval rate with a minimum of 10000 tasks and with a Master rating. To prevent participants from working in more than one posting (condition) of this study, the experimenter assigned an ID to all workers who completed a task, which prevented them from accepting another.

In the 2AFC, thirty-two participants (22 male, 10 female) with an age range of 26-49 (mean=34.75, standard deviation=6.97) were involved in this study. Sixteen participants reported prior robot experience with seven participants reporting robot ownership. Thirteen participants reported prior piloting experience with remote controlled or manned aircraft. Twenty-one participants had completed a bachelor's degree or higher, four completed some college, three had an associate's degree, and four completed high school.

The 7AFC had thirty-two participants (21 male, 11 female) with an age range of 21-61 (mean=33.88, standard deviation=9.06). Seven participants reported prior robot experience with four reporting robot ownership. Thirteen participants reported prior experience piloting a remote controlled or manned aircraft. Fifteen participants completed a bachelor's degree or higher, eleven had some college, two had an associate's degree, and four completed high school.

IV. SUAS FLIGHT PATH DESIGN | INSERTED AFTER SOURCE PAGE 4

How the eight visible gestures are generated

PURPOSE. This section converts qualitative bird-like motions into repeatable parametric trajectories, flies them under motion-capture feedback, and records standardised videos for the human study.

What this section is doing

- 1 Choose seven possible meanings relevant to operators or bystanders: lost signal, lost sensor, draw attention, landing, missed goal, change position, and low battery.
- 2 Select eight periodic shapes: Circle, Figure-8, Left-Right, Loop, Spiral, Swoop, Undulate, and Up-Down.
- 3 Assign an initial developer-proposed meaning to each path before testing human agreement.
- 4 Express $x(t)$, $y(t)$, and $z(t)$ as time functions and use a Vicon-tracked moving target plus PID pose controller to follow them at about 1 m/s.
- 5 Fly each gesture for about 30 seconds and standardise the fixed-camera videos to 854 x 480 pixels with motor sound retained.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Parametric trajectory	A path whose x, y, and z coordinates are explicit functions of time.
Amplitude	The size of an oscillation or maximum displacement from its centre.
Frequency	How quickly a periodic motion repeats.
Phase	The position within a repeating motion cycle.
Vicon	An optical motion-capture system that tracks reflective markers with high position accuracy.
Pose controller	A feedback controller that tracks position and orientation targets.
Orthogonal viewpoint	A camera orientation approximately perpendicular to the main plane of the motion.
Displacement	The spatial extent of the performed gesture along each axis.

How to read the equations, figures, or evidence

- Table I is a reusable shape library. Sine and cosine create smooth periodic paths; multiplying by a shrinking factor creates the spiral.
- The reported displacements are large, roughly 0.5-4 m depending on axis and gesture, and the demonstrations last tens of seconds.
- Vicon and PID ensure the video shows the intended path. They are not the communication receiver used by participants.

What transfers to the EE6008 project

- Implement Circle, Figure-8, Left-Right, and Up-Down as machine-recognition sanity checks and conspicuous positive controls.
- Normalise each path by amplitude and duration, then initially scale candidate micro-motion to roughly 10-20 cm and 1.5-2.5 s subject to controller limits.
- Keep Spiral-to-Landing as an optional overt safety gesture, separate from the keyed payload codebook.

- Use jerk-limited smoothing; direct scaling of a position equation can still create unacceptable velocity or acceleration when frequency changes.

CRITICAL READING. These gestures are deliberately salient and can interrupt the primary mission. Their scale, duration, and fixed viewing plane make them unsuitable as final covert symbols without major redesign.

Check your understanding

- Which parameters control the size and speed of a sinusoidal path?
- Why is a Vicon-controlled trajectory useful even when the final receiver is a camera?

Instructions

On the following pages, there are statements describing feelings about robots. Please use the rating scale below to describe how accurately each statement reflects your feelings. Describe how you generally feel now, not as you wish to feel in the future. So that you can describe yourself in an honest manner, your responses will be kept in absolute confidence. Please read each statement carefully, and then circle the number on the scale.

Response Options

- 1: I Strongly Disagree
- 2: I Disagree
- 3: Undecided
- 4: I Agree
- 5: I Strongly Agree

I would feel uneasy if robots really had emotions.

- select one -

Something bad might happen if robots developed into living beings.

- select one -

Fig. 3. Participant view of the mTurk task.

One important note about the robot experience questions is that they were phrased to solicit interactions in a broad context. Robot experience was assessed by: asking whether participants had “ever interacted with a robot”, the frequency of interaction, and the type of robot (consumer, including Roomba or a pool cleaning robot; industrial, including telepresence or other workplace robots; educational, including Lego Mindstorms or those in a museum; or entertainment, including Parrot AR.drone, DJI Phantom, or Sony Aibo). These prompts also serve to remind people about times when they may have interacted with a robot, so might be overly sensitive to individual interactions. Within the participants that had robot interactions, 6 of 23 had only interacted with a robot once.

C. Experimental Procedure

Participants were required to click a button labeled “I Accept” to consent to participate in the study, otherwise they had to decline and return the task. After accepting, they were given a short demographics survey, positive and negative affect assessment, and negative attitudes towards robots scale to answer. Upon completion of the questions on mTurk, the participants were redirected to a Google Form (while being instructed to keep the task open for a completion phrase to be provided at the end of the Google Form). This allowed the task time to be tracked and all information for task approval to remain within mTurk, while also enabling filtering of repeated workers and a fixed ordering of both the videos and post-experiment questionnaires. While not very streamlined, this guaranteed ordering was important for the counter-balanced conditions in order to prevent the participants from reordering on the single page design required by mTurk (participant view is shown in Fig. 3).

The Google Form began by requiring the mTurk ID to prevent repeated access by the same workers. Next, participants were required to perform a manipulation check to ensure that their computer displayed video and they could follow directions (enter the word displayed in a video similar to those they would watch for the task). Following this, they were shown a set of four videos (counterbalanced in four presentation orders with eight participants each for each experiment). For each video they were shown an embedded video with a label that read “Please use this video to answer the next question.” The task was shown below with the title

Robot Labeling Group A

B

2

Please use this video to answer the next question.

Video Labeling B

Watch the above video and select the word that best describes what the robot is trying to communicate.

☐ Lost Sensor

☐ Landing

« Back Continue »

21% completed

Fig. 4. Participant view of a Google Form question.

of the video and a subtitle reading “Watch the above video and select the word that best describes what the robot is trying to communicate.” and below this was a set of either two words or seven words depending upon the participant condition (a sample 2AFC participant view is shown in Fig. 4). After completing these four videos, another manipulation check was performed before the participant continued to another four videos to label (with each participant seeing all eight videos). Finally, a post-experiment questionnaire was administered to assess: positive and negative affect, whether they thought the robot was looking at them, whether they were scared of the robot and if they would approach it outdoors, comments about the robot, comments about the experiment, and general comments.

Participants completed the entire task in an average of 24.63 minutes (SD=12.18) in the 2AFC task and 26.15 minutes (SD=12.29) in the 7AFC task.

D. Experimental Materials

The same videos were used for all of the experiments, with the only difference in materials being the choices provided in the forced-choice tasks, shown in Table II.

E. Analysis and Results

The results in these studies were judged using a binomial test for 2AFC (compared to 50%) and a Chi-square test (compared to an even distribution) with $p < 0.01$; the resultant necessary agreement was 75% agreement in 2AFC and 34.4% agreement in 7AFC. In the 2AFC, subjects labeled gestures with a chosen label compared with a distractor chosen from a set of seven labels (draw attention, lost sensor, landing, change position, lost signal, low battery, and missed goal). Those labeled with high agreement are: Figure-8 (Lost

TABLE II
AVAILABLE LABELS FOR EACH CONDITION.

Motion	2AFC		7AFC
	Label	Distractor	Label
Circle	Lost Signal	Draw Attention	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Figure-8	Lost Sensor	Landing	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Left-Right	Missed Goal	Lost Sensor	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Loop	Draw Attention	Low Battery	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Spiral	Landing	Missed Goal	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Swoop	Draw Attention	Low Battery	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Undulate	Change Position	Low Battery	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery
Up-Down	Change Position	Lost Signal	Lost Signal, Change Position, Landing, Missed Goal, Draw Attention, Lost Sensor, Low Battery

TABLE III
RESULTS FROM A PRELIMINARY TEST WITH 64 PARTICIPANTS (32 IN EACH CONDITION), WITH ONLY $p < 0.01$ SHOWN.

Motion	2AFC		7AFC	
	Percent Chosen	Label	Percent Chosen	Label
Circle			40.6%	Draw Attention
Figure-8	84.38%	Lost Sensor	40.6%	Change Position
Left-Right				
Loop			34.4%	Landing
Spiral	87.5%	Landing	59.4%	Landing
Swoop	75%	Draw Attention		
Undulate			34.4%	Draw Attention
Up-Down				

Sensor, 84.38%), Spiral (Landing, 87.5%), and Swoop (Draw Attention, 75%). To further explore these results, a 7AFC test was run with all labels available for each motion. These results suggest that 5 motions (3 unique from the first set) could be discriminated at a $p < 0.01$, these are: Circle (Draw Attention, 40.6%), Figure-8 (Change Position, 40.6%), Loop (Landing, 34.4%), Spiral (Landing, 59.4%), and Undulate (Draw Attention, 34.4%). These results are also shown in Table III.

VI. DISCUSSION

We now discuss the results found in the studies presented above before remarking upon the manipulation considerations, limitations, research and design implications, and future work related to these studies.

A. Use of Avian Flight Paths to Convey sUAS State Information

The primary finding in this work was that sUAS flight paths can be used to communicate information about the sUAS and its understanding of the world to diverse participants. Participants broadly agreed that a Spiral could be used to indicate landing and participants in the 7AFC also suggested that Undulate could be used to draw attention, and we believe these gestures could likely be used by developers now to indicate these states in public facing interactions. It was also interesting that there was significant agreement in 2AFC for the use of Figure-8 for lost sensor and Swoop for draw attention, but that Figure-8 was more recognized for change position in the 7AFC. This suggests that users may understand in the presence of suggestion but might be less likely to apply these labels without prompting.

The states proposed in this work were based on a developer's understanding of states that would be helpful to

communicate and which might align with participants' prior experiences in the world. Also of interest is that the assumptions about the participants' likely mental models were largely unconfirmed in the 2AFC study, but the 7AFC study should provide insight into the ability for participants to engage with labels for future studies. When less constrained in their choices, the participants largely agreed about gestures to draw attention and to land, which might be due to the ability of novice users to better understand these states. Participants without experience in working with sUAS may not understand the impact and standard responses to states such as lost signal or lost sensor. This was thought to be a benefit when designing the study, but may have led to a reluctance to use these labels or a less principled application of their use. Draw attention and landing were both used at least 20% of the time, while the others were used 15% or less of the time and lost sensor was only used in 10% of responses across both conditions. A recommendation would be to follow these studies up with an open-ended study, but this would be largely dependent upon the instructions given to the participants and the ability to create meaningful categories from the unconstrained data. On the other hand, it would provide a richer data set from which to generalize states to communicate.

B. Impact of Negative Attitudes towards Robots

Riek et. al [14] found that people with negative attitudes towards robots, as assessed by the Negative Attitudes towards Robots (NARS) scale [31] had more difficulty in recognizing robot gestures when interacting with a humanoid robot. We also examined our data to assess whether this finding was supported with sUAS and the results are presented in Table IV. There were 17 participants with positive attitudes towards robots (8 in 7AFC and 9 in 2AFC with means below 2) and

V. FORCED CHOICE LABELING STUDY | INSERTED AFTER SOURCE PAGE 6

How semantic agreement is measured

PURPOSE. This section tests whether people choose the same meanings for each gesture under two levels of constraint: two possible labels and all seven candidate labels.

What this section is doing

- 1 Recruit 64 qualified Mechanical Turk participants and split them into independent 2AFC and 7AFC groups of 32.
- 2 Show all eight standardised videos, counterbalance four presentation orders, and insert manipulation checks to verify playback and instruction following.
- 3 In 2AFC, compare the developer label with one distractor; in 7AFC, allow all seven labels.
- 4 Use a binomial test against 50% for 2AFC and a chi-square test against equal choice probability for 7AFC, with $p < 0.01$.
- 5 Identify significant gesture-label pairs and compare whether the interpretation remains stable when the choice set changes.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
2AFC / 7AFC	Two- or seven-alternative forced choice: the participant must select from a fixed list.
Distractor	An alternative answer included to test whether the intended label is preferred.
Chance baseline	Expected performance if choices were random: 50% in a two-choice task.
Binomial test	A test of whether the number choosing one option differs from a specified probability.
Chi-square test	A test comparing observed category counts with an expected distribution.
p-value	The probability of results at least this extreme under the null model; it is not the probability the hypothesis is true.
Counterbalancing	Varying presentation order across participants to reduce order effects.
Manipulation check	A task confirming participants can view the stimulus and follow instructions.

How to read the equations, figures, or evidence

- Significant 2AFC pairs are Figure-8/lost sensor 84.38%, Spiral/landing 87.5%, and Swoop/draw attention 75%.
- Significant 7AFC pairs include Circle/draw attention 40.6%, Figure-8/change position 40.6%, Loop/landing 34.4%, Spiral/landing 59.4%, and Undulate/draw attention 34.4%.
- Only Spiral-to-Landing is stable across both choice conditions. Figure-8 changes meaning when the available labels change, showing prompt dependence.
- The thresholds 75% and 34.4% arise from the selected statistical tests and sample size; they are not universal recognition standards.

What transfers to the EE6008 project

- Convert a machine receiver's symbol predictions into a confusion matrix analogous to the human label table.
- Use large gestures to test that Bob recognises signals and Willie detects conspicuous behaviour before evaluating micro-motion.

- For digital communication, replace semantic agreement with BER, PER, synchronisation success, and uncertainty over repeated flights.

CRITICAL READING. Forced choice can inflate apparent understanding, labels were proposed by developers, and the experiment used video rather than live interaction. No bit framing, machine decoder, throughput, or error-correction mechanism is studied.

Check your understanding

- Why is 40.6% potentially significant in 7AFC but weak in 2AFC?
- Which mapping is robust to changing the label set?

TABLE IV

RESULTS FROM A PRELIMINARY TEST WITH 64 PARTICIPANTS (32 IN EACH CONDITION) COMPARING BROAD AGREEMENT OF PARTICIPANTS WITH NEGATIVE ATTITUDES TOWARDS ROBOTS VERSUS THOSE WITH POSITIVE ATTITUDES TOWARDS ROBOTS.

Motion	2FAC			7FAC		
	Positive Attitude	Negative Attitude	Label	Positive Attitude	Negative Attitude	Label
Circle	77.8%	50%	Draw Attention		50%	Draw Attention
Figure 8	77.8%	100%	Lost Sensor	50% Draw Attention	50% Change Position	
Left-Right	66.7%	100%	Lost Sensor	50%		Missed Goal
Loop	77.8%	50%	Draw Attention			
Spiral	88.9%	83.3%	Landing	50%	66.7%	Landing
Swoop	88.9 %	66.7%	Draw Attention	50%	50%	Low Battery
Undulate	55.6%	50%	Change Position			
Up-Down	66.7%	50%	Change Position		50%	Lost Signal

12 participants with negative attitudes towards robots (6 in each 2AFC and 7AFC with means above 3).

This data does not seem to support the Riek finding, which could be due to multiple factors such as the lack of a clear meaning for prejudice against robots to interfere with, lack of mapping from humanoid or ground robots to aerial vehicles, and too small samples for those with extreme opinions about robots. Note when viewing the table that 50% was the expected agreement in the 2AFC, which was shown 4 times in the negative attitudes conditions, but the two highest agreements also appeared in the negative attitude conditions. Across all attitudes and conditions, the recognition of spiral relating to landing was shown.

C. Manipulation Considerations

To confirm that users were able to play the videos and follow directions, each task had two test tasks, which said “Watch the video above and type the word shown.” Each of these videos displayed words on the screen and participants were required to type those words into a text box to have their tasks approved. These test tasks were displayed at the beginning of the task and again halfway through the labeling task.

One manipulation that we did not assess, but should be considered in future work, is whether the users have sound enabled on their computer. This task did not require sound to be enabled, but also did not remove the sound from the videos, which may have resulted in different sensory interactions from different participants. Some participants mentioned the sound in their post-experiment feedback on the robot, for example:

- P14, 2AFC: “It’s very loud. I can see how the noise can become grating soon enough.”
- P11, 7AFC: “Never realized just how loud they are.”
- P15, 7AFC: “It is an excessively loud robot. ”

D. Limitations

At this formative stage, we attempted to control several factors to gain an understanding of communicative ability with limited complexity. The work is replicable and also a valuable starting point that has been successfully used in human-human communication studies [10], [9]. The design choices we made, however, limit the generalization of our findings, and we intend to relax them in the next studies.

Among those are moving from indoor flights to outdoor flights and from video to live flights, adding goal states, and enriching both the response set and the whole context.

From a population perspective, since the participants were recruited from mTurk, they did have more diversity in age and education than we would generally see on a college campus, but they were overwhelmingly American which could limit the application of these results in other cultures, as suggested by [32].

One final limitation of note is the idea that the gestures could interfere with other tasks being undertaken by the sUAS. While this is a possible drawback from the visual communication, it is anticipated that the states proposed for communication would necessitate the pausing of a task. For example, if a sUAS is taking pictures or videos of a farm and finds something important then the sensed information would potentially override the value of the other mission. The states presented here are targeted for generality, but some such as attract attention, may be less valuable to communicate in hobby domains and essential in applied domains.

E. Research and Design Implications

A major implication of this work is to suggest investigations into how naive users understand sUAS problem states when compared to users with experience in piloting these vehicles. The communication of more technical states will be predicated upon a user base that can understand them, which may result in recommendations for team training prior to implementing use. This training will be unlikely with general novice users, but may be resolved through a more participatory design process or through communication of the states they do understand. For example, it may be less important that bystanders understand the “lost sensor” signal if it is followed by a spiral landing, which they can generally understand and predict the final state.

Another path for exploration is the idea of a “guessability” study as conducted by [33], [16], [23] to have a small set of participants design gestures and then test them with a larger pool in order to see if there are understood methods for conveying these ideas. An interesting aspect of these tests is that the users may converge on any number of parameters, such as height change indicating attention drawing and combine these ideas into a richer gesture set than the one that we applied here. This approach has limitations which

were considered to make it an inappropriate starting point (such as the limitation to the culture that created the gestures as seen in [10]) while the labeling study here was deemed to be a sanity check to understand whether opinions could converge on given flight paths. With the information here, it is expected that a “guessability study” would be a valid method for refinement, allowing a comparison against the baseline described here and is recommended for additional study.

Additionally, it is recommended that the Negative Attitudes towards Robots scale [31] be investigated for applicability to sUAS research. This recommendation is based on the inconclusive support for Riek’s [14] humanoid findings to the sUAS gestures.

VII. CONCLUSIONS

This paper presents an initial study on the ability for novice users to understand communicative flight paths from an sUAS to increase safety in future interactions in public spaces. Results indicate a strong understanding across users for a spiraling path to communicate “landing”, but users primarily gravitated towards well understood states (draw attention and landing) while avoiding more technical states (lost sensor). Recommendations for future work include: outdoor tests, visible goal state, open-ended responses, and user generated flight paths. An initial finding questions whether the findings from humanoid robots that negative attitudes towards robots decrease understanding of gestures also applies to sUAS, and leaves a more broad question on whether a version of NARS should be revised to apply specifically to sUAS.

ACKNOWLEDGMENT

The authors thank Urja Acharya, Rubi Quinones, and Siya Kunde for their technical assistance.

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VI. DISCUSSION | INSERTED AFTER SOURCE PAGE 8

Interpreting the human results and their limits

PURPOSE. This section explains why people favour familiar meanings, tests whether robot attitudes matter, and identifies design limitations and future study directions.

What this section is doing

- 1 Interpret landing and attention as familiar, action-oriented states that novices can more readily understand.
- 2 Note that technical labels such as lost sensor receive little support without prompting or domain knowledge.
- 3 Compare participants with strongly positive or negative robot attitudes using the NARS scale, but recognise the extreme subgroups are small.
- 4 Discuss uncontrolled sound, indoor fixed-video presentation, US-heavy recruitment, absence of goals/context, and possible interference with the UAV's main task.
- 5 Recommend open-ended, participatory, and guessability studies for richer gesture design.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
NARS	Negative Attitudes toward Robots Scale, a questionnaire measuring discomfort or negative beliefs about robots.
Generalisation	Whether findings remain valid for new people, cultures, platforms, environments, and tasks.
Ecological validity	How closely an experiment resembles the real situation in which the behaviour will be used.
Participatory design	Involving intended users in creating and refining the interaction design.
Guessability study	Users propose gestures first; a larger group then tests whether those user-generated gestures are interpretable.
Sensory confound	An uncontrolled cue, such as motor sound, that may influence judgments attributed to visual motion.

How to read the equations, figures, or evidence

- The NARS comparison is exploratory because only 17 positive-attitude and 12 negative-attitude participants fall into the extreme subgroups.
- Participant comments about loudness show that some users attended to audio, although the study did not control whether sound was enabled.
- The authors explicitly acknowledge video/live, indoor/outdoor, cultural, context, and mission-interruption limitations.

What transfers to the EE6008 project

- Use machine Willie rather than human subjects for the core project; reserve human salience work as a separately approved extension.
- Model motor sound or remove it consistently if human perception is ever studied, because sound may reveal the gesture.
- Adopt a two-layer design: overt public safety gestures for landing/attention and a separate keyed micro-motion layer for digital payload.
- Test multiple viewpoints, distances, occlusion, lighting, and normal mission contexts rather than assuming geometric shape alone is invariant.

CRITICAL READING. The data support a narrow conclusion: certain large gestures attract common labels in a controlled video task. They do not demonstrate a universal language of drone motion or a robust communication channel.

Check your understanding

- Which limitations threaten cultural and real-world generalisation?
- Why is the NARS conclusion inconclusive?

VII. CONCLUSIONS | INSERTED AFTER SOURCE PAGE 8

What to carry forward from the HRI baseline

PURPOSE. The conclusion identifies Spiral-to-Landing as the clearest public signal and recommends broader future studies. Its project value is a shape library and visibility baseline.

What this section is doing

- 1 Restate that novice observers can attach common meaning to some UAV flight paths.
- 2 Emphasise strong understanding of the spiralling landing path and preference for familiar labels.
- 3 Recommend outdoor/live tests, visible goals, open-ended responses, user-generated paths, and further study of robot-attitude measures.

Professional vocabulary

TERM	PLAIN-ENGLISH MEANING
Overt signal	A deliberately noticeable signal intended to be recognised by observers.
Positive control	A condition expected to produce a clear effect, verifying that the detector or experiment can respond.
Motion alphabet	A set of distinct trajectory shapes treated as candidate symbols.
Public safety layer	An unencrypted, conspicuous behaviour used to warn nearby people of a state such as landing.

How to read the equations, figures, or evidence

- The central robust semantic result is Spiral-to-Landing, not the complete original set of developer assignments.
- Left-Right and Up-Down show no significant consensus in the main result table, so intuitive anthropomorphic analogies are not sufficient evidence.

What transfers to the EE6008 project

- Use the eight trajectories as normalised templates, but label them as overt baselines unless a new detectability test proves otherwise.
- Compare Bob reliability and Willie detectability for original large paths, scaled paths, and keyed mission-embedded micro-motion.
- Report explicitly that semantic understandability and statistical covertness are different objectives and may conflict.

CRITICAL READING. Do not cite this paper as evidence that large motion shapes are covert, high-rate, viewpoint-invariant, or machine-decodable. It is an HRI labelling study.

Check your understanding

- What is the correct role of Ref C in the EE6008 experiment matrix?
- Which gesture could form an overt safety layer, and why?